

AFL Data Foundations — Data Dictionary & Target Definitions

Day-1 contract for Day-2 modelling. All predictive features must use information available strictly before the target match.

1. Prediction targets

Target	Definition / formula	Grain	Task / policy
match_result_3way	HOME_WIN, AWAY_WIN, or DRAW based on the designated home-team result.	Match	3-class classification; draws retained.
margin	home_score – away_score. Positive = home win; negative = away win.	Match	Regression; secondary target.
is_top_disposal_getter	1 when player disposals equals the maximum valid disposals in that match; otherwise 0.	Player-match	Binary; ties retained; all-missing metric = NA.
is_top_goal_kicker	1 when player goals equals the maximum valid goals in that match; otherwise 0.	Player-match	Binary; ties retained; all-missing metric = NA.
custom_performance_score	3K + 2HB + 3M + 4T + 6G + B + HO + FK_for – 3FK_against + 5CL – 2CLG.	Player-match	Continuous diagnostic; not official AFL Fantasy scoring.
is_top_custom_performance	1 when custom_performance_score equals the match maximum; otherwise 0.	Player-match	Binary; ties retained; all-missing metric = NA.

2. Match-level predictive features

Feature family	Definition / computation	Window / leakage rule
Form	form_win_rate_last3/5/10; avg_score_last3/5/10; avg_conceded_last3/5/10	Prior N matches only: shift(1) → rolling(N).
Win streak	Consecutive prior wins entering the match.	Prior results only; current result excluded.
Head-to-head	h2h_win_rate_prior and h2h_games_played_prior by team/opponent.	Prior meetings only: shift(1) → expanding.
Venue history	venue_win_rate_prior by team/venue.	Prior matches at that venue only.
Rest	Days since previous team match.	Strict team chronological order.
Season form	season_win_rate_prior.	Prior games in current season only.
Ladder proxy	ladder_rank_prior based on prior win rate, not official AFL ladder position.	Only games/results before match date; same-day results excluded.
Travel / venue	is_interstate and is_neutral_venue.	State mappings are explicit; unknown interstate mappings remain NA.

3. Player predictive features

For each player, last-3, last-5 and last-10 averages are calculated for disposals, goals and supplied fantasy_points using shift(1) before rolling. The current match statistic is therefore never used in its own predictor.

4. Data-quality and evaluation contract

Exact duplicate player-game rows are removed; invalid negative disposals/fantasy_points are excluded using the union of invalid rows; high positive values are retained unless independently impossible. Missing historical statistics are preserved rather than

automatically converted to zero. The final complete season is held out as future data, with all earlier seasons used for training. A random split is avoided because it can place future matches in training data.

Day-2 rule: imputation, scaling, feature selection and model tuning must be fitted on training data only and then applied unchanged to the hold-out.